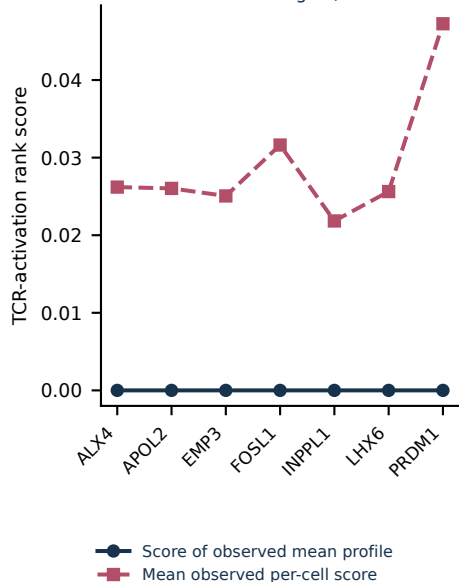


a Aggregation changes the target

Schmidt: all 7 held targets; observed data only



b OP3 program concordance

FP-ridge *	0.81	0.28 2/4	0.20 1/4
biolord	0.52	0.25 2/4	-0.20 2/4
CellFlow	0.65	NA	-0.29 1/4
CellIOT	NA-P	NA	NA
PRnet	0.51	NA	-0.14 1/4
scPRAM	0.07 3/4	NA	-0.05 1/4
scFoundation †	NA-P	NA	NA
scGPT †	NA-P	NA	NA
	Type-I IFN	NF-κB	Effector (lymphocyte)

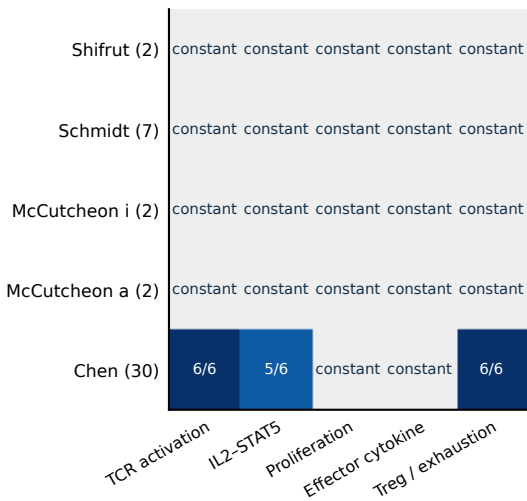
Mean over the lineages where the score is defined;
n/4 printed where fewer than four.

NA-O: constant observed target; NA-P: constant
prediction; NA: both, in different lineages.

Never plotted as zero.

† Adapted * Diagnostic comparator

c T3 estimability across 6 methods



Rows: dataset-arms (held-gene counts in parentheses).
22/25 targets are constant; not evidence of model failure.

d Checkpoint assay context

Frangieh protein fit; 124 held KOs

